

Unit II: SQL Syntax

What You Will Learn

1. Basic syntax regarding how to read, create, modify, and remove data from databases.
2. Practical applications and modifications of the syntax for real-world scenarios.

Basic Syntax Overview

SQL (Structured Query Language) follows a declarative structure. To understand the relationship between keywords and identifiers, consider the following query: `SELECT 'id', 'name' FROM 'new_schema`.`users`;`

In this statement, **SELECT** and **FROM** are the primary SQL keywords. The identifiers `id` and `name` represent column names, while `new_schema` and `users` identify the schema and table respectively.

SQL Syntax Highlights

- **INSERT:** Adding new records
- **SELECT:** Retrieving data
- **UPDATE:** Modifying existing records
- **DELETE:** Removing records
- **WHERE:** Filtering data based on conditions
- **NULL VALUES:** Managing missing information
- **IN, BETWEEN & LIKE:** Advanced filtering patterns

Sample Dataset

Sample data

Let's start with a sample table **users**, to help you understand what kind of results will be presented when an SQL statement is executed.

id	name	age
1	John	50
2	May	40
3	Tim	10

To help solidify the new concepts, you can follow along by editing the provided table below:

```
Schema SQL Query SQL Results E
CREATE SCHEMA `new_schema` DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

CREATE TABLE `new_schema`.`users` (
  `id` INT NOT NULL AUTO_INCREMENT COMMENT 'This is the primary index',
  `name` VARCHAR(45) NOT NULL DEFAULT 'N/A',
  `age` INT NULL,
  `height` INT NULL,
  PRIMARY KEY (`id`)
);

INSERT INTO `new_schema`.`users` (`id`, `name`, `age`, `height`) VALUES (1, 'John', 50, 180);
INSERT INTO `new_schema`.`users` (`id`, `name`, `age`) VALUES (2, 'May', 40);
INSERT INTO `new_schema`.`users` (`id`, `name`, `age`, `height`) VALUES (3, 'Tim', 10, 170);
INSERT INTO `new_schema`.`users` (`id`, `name`, `age`, `height`) VALUES (4, 'Jay', 20, 155);
```

1. Create Data: INSERT

The **INSERT INTO** statement adds new records. It is followed by the table name, the target columns, and the **VALUES** keyword containing the data.INSERT INTO users (id, name, age) VALUES (4, 'Harry', 33);

Application: This adds a new user named Harry, aged 33, to the database.

2. Read Data: SELECT

The **SELECT** clause determines which columns are displayed, while **FROM** specifies the source table.SELECT id, name FROM users;

To retrieve all columns from a table, use the asterisk wildcard:SELECT * FROM users;

Note: For performance optimization in large databases, it is recommended to select only the specific columns required for your task.

3. Modify Data: UPDATE

UPDATE is used to change existing values. It is paired with **SET** to define the new data and **WHERE** to target specific rows. UPDATE users SET name = 'Andy', age = 100 WHERE id = 2;

Warning: Neglecting the **WHERE** clause in an UPDATE statement will apply the changes to every single record in the table.

4. Remove Data: DELETE

The **DELETE** statement removes records. Much like the update statement, it relies heavily on the **WHERE** clause for precision. DELETE FROM users WHERE id = 1;

Caution: Always verify your WHERE condition before executing a deletion to prevent accidental data loss.

5. WHERE: Filter Data

The **WHERE** clause acts as a filter, ensuring only rows that satisfy specific criteria are returned or acted upon. SELECT * FROM users WHERE age > 30;

Common operators include: =, >, <, and !=. Logical operators can also be combined: SELECT * FROM users WHERE age < 40 AND height > 160;

6. Null Values

In SQL, **NULL** represents missing or unknown data. It cannot be evaluated using standard equality operators. Instead, use **IS NULL** or **IS NOT NULL**. SELECT * FROM users WHERE height IS NULL;

7. IN, BETWEEN & LIKE

These operators provide flexible ways to query ranges and patterns:

- **IN:** Matches a value against a specific list.
SELECT * FROM users WHERE id IN (1, 3);
- **BETWEEN:** Filters results within a specified inclusive range.
SELECT * FROM users WHERE age BETWEEN 20 AND 40;
- **LIKE:** Performs pattern matching. The % wildcard matches any number of characters.
SELECT * FROM users WHERE name LIKE '%a%';

Movie Database Management Case Study

1. List movies released in the year 2020.

```
select * from movies where rel_year = 2020;
```

	rel_year	mov_name	act_name	lang	dir_name
►	2020	Big Brother	Mohanlal	Malayalam	Siddique
	2020	Darbar	Rajinikanth	Tamil	AR Murugadoss
	2020	Darbar	Nayanthara	Tamil	AR Murugadoss
	2020	Karthik Dial Seytha Yenn	Silambarasan	Tamil	Gautham Menon
	2020	Karthik Dial Seytha Yenn	Trisha	Tamil	Gautham Menon
	2020	Maanaadu	Silambarasan	Tamil	Venkat Prabhu
	2020	Marakkar: Arabikadalinte Simham	Mohanlal	Malayalam	Priyadarshan
	2020	Pattas	Dhanush	Tamil	R. S. Durai Senthilkumar
	2020	Jagame Thandhiram	Dhanush	Tamil	Karthik Subbaraj
	2020	Varane Avashyamund	Dulquer Salmaan	Malayalam	
	2020	Kannum Kannum Kollaiyadithaal	Dulquer Salmaan	Tamil	
	2020	Kurup	Dulquer Salmaan	Malayalam	
	2020	Sarileru Neekevvaru	Rashmika Mand...	Telugu	
	2020	Bheeshma	Rashmika Mand...	Telugu	
	2020	Pogaru	Rashmika Mand...	Kannada	
	2020	Sultan	Rashmika Mand...	Tamil	
	2020	Pushpa	Rashmika Mand...	Telugu	
	2020	Marakkar: Arabikadalinte Simham	Manju Warior	Malayalam	
	2020	Jack and Jill	Manju Warior	Malayalam	
	2020	Love Story	Sai Pallavi	Telugu	
	2020	Viraata Parvam	Sai Pallavi	Telugu	
	2020	Rachiyamma	Parvathy Thiruv...	Malayalam	
	2020	Halal Love Story	Parvathy Thiruv...	Malayalam	
	2020	Oh My Kadavule	Vijay Sethupathi	Tamil	Ashwath Marimuthu
	2020	Kadaisi Vivasayi	Vijay Sethupathi	Tamil	M. Manikandan
	2020	Uppena	Vijay Sethupathi	Tamil	Buchi Babu Sana
	2020	Master	Vijay Sethupathi	Tamil	Lokesh Kanagaraj
	2020	Maamanithan	Vijay Sethupathi	Tamil	Seenu Ramasamy
	2020	Laabam	Vijay Sethupathi	Tamil	S. P. Jananathan

2. Retrieve movies acted by a specific actor (e.g., “Rajini”).

```
select mov_name from movies where act_name = "Jayam Ravi";
```

	mov_name
▶	Bava Bavamaridi
	Palnati Pourusham
	Jayam
	M. Kumaran S/O Mahalakshmi
	Daas
	Mazhai
	Idhaya Thirudan
	Unnakum Ennakum
	Deepavali
	Velli Thirai
	Santhosh Subramaniyam
	Dhaam Dhoom
	Peranmai
	Thillalangadi
	Ko
	Engenyum Kadhal
	Aadhi Bhagavan
	Ninaithathu Yaaro
	Nimirndhu Nil
	Janda Pai Kapi Raju
	Romeo Juliet
	Sakalakala Vallavan
	Thani Oruvan
	Bhooloham
	Miruthan
	Bogan
	Vanamagan
	Tik Tik Tik
	Adanqa Maru

3. Display movies directed by a particular director.

```
select * from movies where dir_name = "K. Shankar";
```

	rel_year	mov_name	act_name	lang	dir_name
▶	1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
	1979	Mangala Vaathiyam	Kamal Haasan	Tamil	K. Shankar

4. List all movies in the Tamil language.

```
select mov_name from movies where lang = "Tamil";
```

	mov_name
▶	Youth
	Yennai Arindhaal
	Yennai Arindhaal
	Yaaradi Nee Mohini
	Yaaradi Nee Mohini
	Yaar?
	Yaadhum Oore Yaavarum Kelir
	Vivegam
	Viswasam
	Viswasam
	Vishwaroopam II
	Vishwaroopam
	Vishwa Thulasi
	Vishnu
	Virumaandi
	Vinnaithaandi Varuvaayaa
	Vinnaithaandi Varuvaayaa
	Villu
	Villu
	Villain
	Vikram Vedha
	Vikram
	Vikadakavi
	Viduthalai
	Vettaiyaadu Vilaiyaadu
	Vettaikaaran
	Vettai
	Vetri Vizha
	Vetri

5. Find movies whose names start with 'K'.

```
select mov_name from movies where mov_name like "k%";
```

	mov_name
▶	Kalathur Kannamma
	Kannum Karalum
	Kurathi Magan
	Kaalachakram
	Kanyakumari
	Katha Sangama
	Kumaara Vijayam
	Kuttavum Shikshayum
	Kabita
	Kavikkuyil
	Kokila
	Kumkuma Rakshe
	Kaathirunna Nimisham
	Kiladi Kittu
	Kalyanaraman
	Kuppathu Raja
	Kaali
	Kaali
	Kadal Meengal
	Kazhugu
	Kaliyamardanam
	Kelkkaatha Sabdham
	Keni
	Komaram
	Kurukkante Kalyanam
	Kattaruvi
	Kattathe Kilikkoodu
	Kinnaram
	Kodunkattu

6. Display movies released between 2010 and 2020.

```
select mov_name from movies where rel_year between 2010 and 2020;
```

	mov_name
▶	Aasal
	Aayirathil Oruvan
	Adhurs
	Alexander the Great
	Best Actor
	Best of Luck
	Bodyguard
	Boss Engira Bhaskaran
	Drona 2010
	Elektra
	Enthiran
	Four Friends
	Goa
	Goa
	Janakan
	Kandahar
	Khatta Meetha
	Kutty Srank
	Manmadan Ambu
	Manmadan Ambu
	Manmadan Ambu
	Naan Mahaan Alla
	Namo Venkatesa
	Oru Naal Varum
	Paiyaa
	Pokkiri Raja
	Pramaani

Filtering & Sorting Demonstration & Explanation:

- Sort movies by release year in ascending order.

```
select mov_name from movies order by rel_year;
```

	mov_name
▶	Kalathur Kannamma
	Kannum Karalum
	Paadha Kaanikkai
	Paarthaal Pasi Theerum
	Anandha Jodhi
	Vanambadi
	Maanavan
	Annai Velankanni
	Anubhavangal Paalichakal
	Kurathi Magan
	Arangetram
	Kaalachakram
	Sollathaan Ninaikkiren
	Anbu Thangai
	Aval Oru Thodar Kathai
	Gumasthavin Magal
	Kanyakumari
	Naan Avanillai
	Panathukkaga
	Paruva Kaalam
	Vishnu Vijayam
	Aayirathil Oruthi
	Andharangam
	Apoorva Raagangal
	Apoorva Raagangal
	Cinema Paithiyam
	Maalai Sooda Vaa
	Mattoru Seetha
	Melnaattu Marumagal

8. Sort movies alphabetically by name.

```
select mov_name from movies order by mov_name;
```

	mov_name
▶	100 Days of Love
	16 Vayathinile
	16 Vayathinile
	1921
	1971: Beyond Borders
	1983
	2
	2 Penkuttikal
	24
	3
	44044
	44058
	5 Sundarikal
	5 Sundarikal (Kullante Bharya segment)
	7aum Arivu
	90ML
	96
	96
	Aa Divasam
	Aa Divasam
	Aa Neram Alppa Dooram
	Aa Rathri
	Aadai
	Aadaminte Vaariyellu
	Aadavari Matalaku Arthale Verule
	Aadhavan
	Aadhavan
	Aadhi
	Aadhi Bhagavan

9. Retrieve the latest 10 movies based on release year.

```
select mov_name,rel_year from movies order by rel_year desc limit 10;
```

	mov_name	rel_year
▶	Rocketry: The Nambi Effect	2022
	Soorai Pottru	2021
	Ram	2021
	Ponniyin Selvan	2021
	Drishyam 2	2021
	Karthik Dial Seytha Yenn	2020
	Maanaadu	2020
	Big Brother	2020
	Karthik Dial Seytha Yenn	2020
	Darbar	2020

10. List the earliest 10 movies.

```
select mov_name,rel_year from movies order by rel_year limit 10;
```

	mov_name	rel_year
▶	Kalathur Kannamma	1960
	Kannum Karalum	1962
	Paadha Kaanikkai	1962
	Paarthaal Pasi Theerum	1962
	Anandha Jodhi	1963
	Vanambadi	1963
	Maanavan	1970
	Annai Velankanni	1971
	Anubhavgal Paalichakal	1971
	Kurathi Magan	1972

11. Find movies where the actor name contains “Khan”.

```
select mov_name,act_name from movies where act_name like "%Haasan%";
```

	mov_name	act_name
▶	Kalathur Kannamma	Kamal Haasan
	Kannum Karalum	Kamal Haasan
	Paadha Kaanikkai	Kamal Haasan
	Paarthaal Pasi Theerum	Kamal Haasan
	Anandha Jodhi	Kamal Haasan
	Vanambadi	Kamal Haasan
	Maanavan	Kamal Haasan
	Annai Velankanni	Kamal Haasan
	Kurathi Magan	Kamal Haasan
	Arangetram	Kamal Haasan
	Sollathan Ninaikkiren	Kamal Haasan
	Anbu Thangai	Kamal Haasan
	Aval Oru Thodar Kathai	Kamal Haasan
	Gumasthavin Magal	Kamal Haasan
	Kanyakumari	Kamal Haasan
	Naan Avanillai	Kamal Haasan
	Panathukkaga	Kamal Haasan
	Paruva Kaalam	Kamal Haasan
	Vishnu Vijayam	Kamal Haasan
	Aayirathil Oruthi	Kamal Haasan
	Andharangam	Kamal Haasan
	Apoorva Raagangal	Kamal Haasan
	Cinema Paithiyam	Kamal Haasan
	Maalai Sooda Vaa	Kamal Haasan
	Mattoru Seetha	Kamal Haasan
	Melnaattu Marumagal	Kamal Haasan
	Njan Ninne Premikkunnu	Kamal Haasan
	Pattampoochi	Kamal Haasan
	Pattikkaattu Raja	Kamal Haasan